

TrES-3b

R-0029

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_260508 · created 2026-07-08T16:38:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2461168.7804 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.057
Transit depth (Rp/Rs)^2	3.79 +/- 2.23 [%]
Orbital Inclination (inc)	86.85 +/- 2.9
Airmass coefficient 1 (a1)	1.24 +/- 0.73
Airmass coefficient 2 (a2)	-0.1 +/- 1.01
Scatter in the residuals of the lightcurve fit is	58.51 %
Best Comparison Star	#1 - [432, 153]
Optimal Aperture	1.11
Optimal Annulus	3.5
Transit Duration (day)	0.077 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.057 vs 0.16309 (deviation 0.56σ)
Duration fit vs published	0.077 d vs 0.0567 d (deviation 1.85σ)
Mid-time O–C	-4.6 min
Depth SNR	3.4
Residual scatter	58.51 %

Light curve

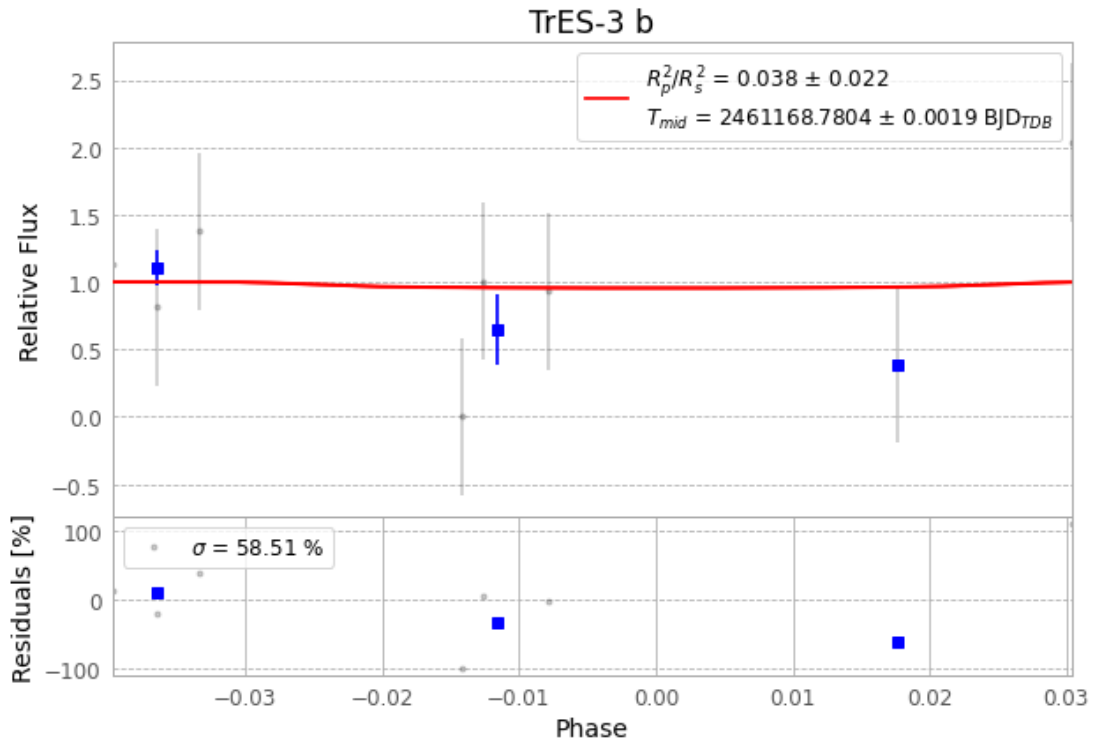


Figure 1 — FinalLightCurve_TrES-3 b_2026-05-07.png (SHA-256 5c455738bb148801...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [266, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1bd8aeed0673abcf...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

66 FITS frames (43 MB), TRES-3260508051100.FITS → TRES-3260508082715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-260508.log (7c9e9ca013fe...), FinalLightCurve_TrES-3 b_2026-05-07.png (5c455738bb14...), FinalLightCurve_TrES-3 b_2026-05-07.csv (2105ac0d8fab...)

Compute resources

Wall time 161.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.